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EE1060: Differential Equations and Transform Techniques

Lecture 33: Laplace Transform and Differential Equations

Topics :

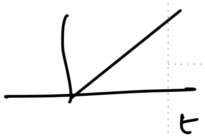
1. Using Laplace Transform to solve ODE
2. Inverse Laplace Transform

Laplace transform - Review

Laplace Transform of Some Common Signals

if it corresponds to $h(t)$ of an LTI System

Signal	LT	ROC	Poles / Zero's	Stability
$\delta(t)$	1	entire s-plane.		
$u(t)$	$1/s$	$\sigma > 0$	Pole at $s=0$	marginally Stable.
$e^{at}u(t)$ $(a>0)$	$1/(s-a)$	$\sigma > a$	Pole at $s=a$	unstable.
$e^{-at}u(t)$ $(a>0)$	$1/(s+a)$	$\sigma > -a$	Pole at $s=-a$	Stable.
$t\,u(t)$	$1/s^2$	$\sigma > 0$	Repeated poles at $s=0$	unstable.
$t^n u(t) \rightarrow$				
$\cos\omega_0 t\; u(t)$	$s/(s^2+\omega_0^2)$	$\sigma > 0$	Poles at $s=\pm j\omega_0 \rightarrow$ sinusoidal S.S; Zero at $s=0$	
$e^{-at}\cos\omega_0 t\; u(t)$	$\dfrac{s+a}{(s+a)^2+\omega_0^2}$	$\boxed{\sigma > -a}$	Poles at $s=-a\pm j\omega_0$; Zero's at $s=-a$ $\rightarrow$	Stable.
$\sin\omega_0 t\; u(t)$	$\omega_0/(s^2+\omega_0^2)$	$\sigma > 0$	Poles at $s=\pm j\omega_0$	



Properties of Laplace Transform.

$$\mathcal{L}\{e^{at}x(t)\} = X(s-a)$$

$$\mathcal{L}\{x(t-\tau)\} = e^{-s\tau}X(s)$$

$$\mathcal{L}\{x^{(n)}(t)\} = s^n X(s) - \underbrace{s^{n-1}x(0) - s^{n-2}x'(0) - \cdots - x^{(n-1)}(0)}_{\text{initial Conditions}}$$

Laplace Transform and Differential Equations

$$\frac{dy}{dt} + a\,y = x(t)$$

apply LT on both sides.

$$s\,Y(s) - y(0) + a\,Y(s) = X(s)$$

$$Y(s)[s+a] - y(0) = X(s)$$

$$Y(s) = \frac{X(s)}{s+a} + \frac{y(0)}{s+a}$$

input in 's' domain

output of the system in 's' domain

(or) Sol of DE

if $y(0) = 0$, $\frac{Y(s)}{X(s)} = \frac{1}{s+a}$

$$\frac{d^2y}{dt^2} + 2\xi\omega_n\frac{dy}{dt} + \omega_n^2 y = x(t)$$

in s-domain:

$$s^2Y(s) - s\,y(0) - y'(0) + 2\xi\omega_n(s\,Y(s) - y(0)) + \omega_n^2\,Y(s) = X(s)$$

$$Y(s) = \frac{X(s)}{s^2+2\xi\omega_n s+\omega_n^2} + \frac{y'(0)}{s^2+2\xi\omega_n s+\omega_n^2} + \frac{y(0)(s+2\xi\omega_n)}{s^2+2\xi\omega_n s+\omega_n^2}$$

if $y(0)=0$ & $y'(0)=0$, $\frac{Y(s)}{X(s)} = \frac{1}{s^2+2\xi\omega_n s+\omega_n^2}$

$H(s) =$

Steps :- given an ODE (time-domain) $\xrightarrow{LT}$ Relation b/w input & o/p in 's' domain (algebraic relation) $\longrightarrow$ Solve for $Y(s)$ $\xrightarrow[\text{Transform}]{\text{inverse}}$ $y(t)$

Inverse Laplace Transform - Partial Fractions

decompose $H(s)$ into factors for which LT is easily recognisable.

$$H(s) = \frac{N(s)}{D(s)} \quad \text{when order of } N(s) < \text{order of } D(s)$$

$$(m) < (n)$$

Simple linear factors:- $H(s) = \dfrac{N(s)}{\text{Prod of } (s+P_i)} = \sum \dfrac{A_i}{s+P_i}$

$[\text{i.e., } D(s) = \prod (s+P_i)]$ $\qquad h(t) = A_i e^{-k_i t} u(t)$

Repeated linear factors:- $H(s) = \dfrac{N(s)}{\text{Prod of } (s+P_i)^k}$ at least for one i, $\underline{\underline{k > 1}}$

Example :- $H(s) = \dfrac{N(s)}{(s+P_1)^2 (s+P_2)^3 (s+P_3)}$

The corresponding PF decomposition is

Eq. time-domain $= t e^{-P_2 t}$

$$H(s) = \frac{A_1}{s+P_1} + \frac{A_2}{(s+P_1)^2} + \frac{A_3}{(s+P_2)} + \frac{A_4}{(s+P_2)^2} + \cdots$$